

```
import streamlit as st
import pandas as pd
import numpy as np
import plotly.graph_objects as go
import time
from datetime import datetime
```

```
# --- 1. PAGE CONFIG & THEME VARIABLES ---
```

```
st.set_page_config(layout="wide", page_title="Smart Air Quality Dashboard")
```

```
# Theme colors
```

```
is_dark = True
```

```
bg_color = "#000000"
```

```
text_color = "#ffffff"
```

```
card_bg = "#1e1e1e"
```

```
card_shadow = "rgba(79,112,255,0.4)"
```

```
alert_bg = "rgba(230, 57, 70, 0.1)"
```

```
# --- 2. GLOBAL STYLES & LOADER ANIMATIONS ---
```

```
st.markdown(f"""
```

```
<style>
```

```
/* Dashboard Layout Tweaks */
```

```
.stApp {{ background-color: {bg_color}; color: {text_color}; }}
```

```
header[data-testid="stHeader"] {{
  background-color: #000000 !important;
}}
```

```
[data-testid="stSidebar"] {{ background-color: #111111 !important; border-right: 1px solid
#333; }}
```

```
[data-testid="column"] {{ background-color: {card_bg} !important; padding: 25px !important;
border-radius: 12px !important; border: 1px solid #333 !important; }}
h1, h3, .stMarkdown p {{ color: {text_color} !important; }}
```

```
/* EARTH LOADER CSS */
```

```
.earth-wrapper {{
  display: flex; flex-direction: column; align-items: center; justify-content: center; height: 80vh;
}}
```

```
.earth-loader {{
  --watercolor: #3f51d9; --landcolor: #9be24f;
  width: 10em; height: 10em; position: relative; overflow: hidden; border-radius: 50%;
  border: 2px solid rgba(255,255,255,0.9);
  background: radial-gradient(circle at 30% 30%, #6a78ff, var(--watercolor));
```

```

    box-shadow: inset 0.45em 0.45em rgba(255,255,255,0.22), inset -0.6em -0.6em
    rgba(0,0,0,0.42), 0 0 30px rgba(79,112,255,0.4);
  }}
  .earth-loader svg {{ position: absolute; width: 10.2em; opacity: 0.9; filter: drop-shadow(0 0 4px
  rgba(155,226,79,0.65)); }}
  .earth-loader svg:nth-child(1) {{ top: -2.6em; animation: round1 4s infinite linear; }}
  .earth-loader svg:nth-child(2) {{ bottom: -2.8em; animation: round2 4s infinite linear 0.9s; }}
  .earth-loader svg:nth-child(3) {{ top: -1.8em; animation: round1 4s infinite linear 1.8s; }}

@keyframes round1 {{
  0% {{ left: -3.5em; transform: rotate(0deg); opacity: 1; }}
  45% {{ left: -8em; transform: rotate(20deg); }}
  46% {{ opacity: 0; }}
  55% {{ left: 9em; transform: rotate(-20deg); }}
  65% {{ opacity: 1; }}
  100% {{ left: -3.5em; transform: rotate(0deg); }}
}}
@keyframes round2 {{
  0% {{ left: 5.5em; transform: rotate(0deg); opacity: 1; }}
  65% {{ left: -9em; transform: rotate(-20deg); }}
  66% {{ opacity: 0; }}
  75% {{ left: 10em; transform: rotate(20deg); }}
  85% {{ opacity: 1; }}
  100% {{ left: 5.5em; transform: rotate(0deg); }}
}}

/* DASHBOARD ANIMATIONS */
.float-title {{ animation: float 3s ease-in-out infinite; }}
@keyframes float {{ 0%, 100% {{ transform: translateY(0); }} 50% {{ transform:
translateY(-6px); }} }}
</style>
""", unsafe_allow_html=True)

```

--- 3. SIDEBAR & ADMIN LOGIC ---

```

if 'main_data' not in st.session_state:
    st.session_state.main_data = pd.DataFrame()

```

with st.sidebar:

```

    st.markdown("### ⚙️ Controls")
    st.selectbox("Monitoring Station", ["Downtown", "East Industrial"])

```

ADDED SELECTORS

```

st.selectbox("Time Range", ["Last 24 Hours", "Last 7 Days", "Last 30 Days"])
st.selectbox("Pollutant", ["PM2.5", "PM10", "NO2", "O3", "SO2"])

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st.selectbox("Forecast Horizon", ["12 Hours", "24 Hours", "48 Hours", "72 Hours"])

st.write("")
st.button("🔄 Update Dashboard")

admin_mode = st.toggle("Admin Mode")

if admin_mode:
    st.markdown("---")
    st.markdown("### 📁 Upload Data for Dashboard")
    uploaded_file = st.file_uploader("Upload Air Quality CSV", type=["csv"])
    if uploaded_file:
        st.session_state.main_data = pd.read_csv(uploaded_file)
        st.success("Data Loaded Successfully!")

# --- 4. SCREEN LOGIC (LOADER -> DASHBOARD) ---
if st.session_state.main_data.empty:
    st.markdown(f"""
    <div class="earth-wrapper">
    <div class="earth-loader">
    <svg viewBox="0 0 200 200"><path fill="#9be24f" d="M100 35 C138 38, 162 68, 158 105
    C154 142, 120 160, 100 156 C62 152, 38 125, 42 100 C46 70, 70 40, 100 35Z"/></svg>
    <svg viewBox="0 0 200 200"><path fill="#9be24f" d="M100 45 C132 48, 152 78, 148 108
    C144 138, 118 148, 100 145 C68 142, 48 120, 52 100 C56 78, 72 50, 100 45Z"/></svg>
    <svg viewBox="0 0 200 200"><path fill="#9be24f" d="M100 40 C130 44, 150 72, 146 104
    C142 136, 118 148, 100 144 C70 140, 50 118, 54 100 C58 74, 74 46, 100 40Z"/></svg>
    </div>
    <h2 style="color:white; margin-top:20px; font-family: 'Segoe UI';">Connecting to Smart
    Aware System...</h2>
    <p style="color:#aaa;">Please enable Admin Mode and upload a CSV in the sidebar to enter
    dashboard</p>
    </div>
    """, unsafe_allow_html=True)
else:
    # DASHBOARD SCREEN
    main_data_df = st.session_state.main_data
    st.markdown(f"<h1 class='float-title' style='text-align:center; margin-bottom: 40px;'>🌍 Smart
    Air Quality Dashboard</h1>", unsafe_allow_html=True)

    current_aqi = main_data_df['AQI'].iloc[-1] if 'AQI' in main_data_df.columns else 164.6

    if current_aqi > 150:
        status_text, status_color, alert_msg = "Unhealthy", "#e63946", "🔴 Unhealthy air quality
        today!"

```

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elif current_aqi > 100:
    status_text, status_color, alert_msg = "Moderate", "#f1c40f", "⚠️ Moderate air quality
detected."
else:
    status_text, status_color, alert_msg = "Good", "#2ecc71", "✅ Air quality is currently Good."

# TOP ROW
col1, col2 = st.columns(2)
with col1:
    st.subheader("Current Air Quality")
    fig_gauge = go.Figure(go.Pie(values=[15, 15, 15, 15, 40], hole=0.75,
        marker_colors=['#a32cc4', '#e63946', '#f1c40f', '#2ecc71', '#333333'],
        textinfo='none', sort=False))
    fig_gauge.add_annotation(text=f"<b>{current_aqi}</b>", x=0.5, y=0.6, font_size=50,
showarrow=False, font_color="white")
    fig_gauge.add_annotation(text=f"AQI<br><span
style='color:{status_color}'>{status_text}</span>", x=0.5, y=0.35, font_size=16,
showarrow=False)
    fig_gauge.update_layout(height=350, margin=dict(t=0,b=0,l=0,r=0), showlegend=False,
paper_bgcolor='rgba(0,0,0,0)')
    st.plotly_chart(fig_gauge, use_container_width=True)

with col2:
    st.subheader("PM2.5 Forecast")
    fig_forecast = go.Figure()
    fig_forecast.add_trace(go.Scatter(x=["00:00", "04:00", "08:00", "12:00", "16:00", "20:00"],
y=[42, 38, 45, 52, 48, 44], name='Historical', line=dict(color='#5c7cfa', width=3)))
    fig_forecast.add_trace(go.Scatter(x=["00:00", "04:00", "08:00", "12:00"], y=[40, 40, 43, 48],
name='Forecast', line=dict(color='#ff8787', width=3, dash='dash'))))
    fig_forecast.update_layout(height=350, margin=dict(t=20,b=20,l=20,r=20),
paper_bgcolor='rgba(0,0,0,0)', plot_bgcolor='rgba(0,0,0,0)', font=dict(color="white"),
legend=dict(orientation="h", x=0.3, y=-0.2))
    st.plotly_chart(fig_forecast, use_container_width=True)

# BOTTOM ROW
col3, col4 = st.columns(2)
with col3:
    st.subheader("Pollutant Concentrations (µg/m³)")
    times = ["00:00", "04:00", "08:00", "12:00", "16:00", "20:00", "24:00"]
    fig_conc = go.Figure()
    fig_conc.add_trace(go.Scatter(x=times, y=[15, 18, 20, 25, 22, 19, 17], name="PM2.5",
line=dict(color='#f9a825', width=2)))
    fig_conc.add_trace(go.Scatter(x=times, y=[30, 32, 35, 40, 38, 32, 30], name="PM10",
line=dict(color='#1976d2', width=2)))

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fig_conc.add_trace(go.Scatter(x=times, y=[20, 22, 28, 35, 30, 25, 22], name="O3",
line=dict(color='#43a047', width=2)))
fig_conc.add_shape(type="line", x0=times[0], y0=42, x1=times[-1], y1=42,
line=dict(color="Red", width=2, dash="dot"))
fig_conc.add_annotation(x=times[-1], y=42, text="WHO Limit", showarrow=False,
yshift=10, font=dict(color="white"), bgcolor="Red")
fig_conc.update_layout(height=350, paper_bgcolor='rgba(0,0,0,0)',
plot_bgcolor='rgba(0,0,0,0)', font=dict(color="white"), legend=dict(orientation="h", x=0.2, y=-0.3))
st.plotly_chart(fig_conc, use_container_width=True)

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with col4:

```

st.subheader("Alert Notifications")
st.markdown(f"""
    <div style="background-color: {alert_bg}; padding: 15px; border-radius: 8px;
margin-bottom: 12px; border-left: 5px solid {status_color};">
        <span style="color: {status_color}; font-weight: bold;">{alert_msg}</span><br>
        <span style="color: #ccc; font-size: 0.85rem;">Current AQI: {current_aqi}. Take
necessary precautions.</span>
    </div>
    <div style="background-color: rgba(253, 126, 20, 0.1); padding: 15px; border-radius:
8px; margin-bottom: 12px; border-left: 5px solid #fd7e14;">
        <span style="color: #fd7e14; font-weight: bold;">⚠️ Trend Warning</span><br>
        <span style="color: #ccc; font-size: 0.85rem;">Levels have increased by
{round(current_aqi * 0.05, 1)}% in the last 4 hours.</span>
    </div>
    """, unsafe_allow_html=True)

```